



Time-Temperature Analysis, Tension and Compression in PBXs

Darla Graff Thompson, Racci Deluca & Geoff W. Brown

To cite this article: Darla Graff Thompson, Racci Deluca & Geoff W. Brown (2012) Time-Temperature Analysis, Tension and Compression in PBXs, Journal of Energetic Materials, 30:4, 299-323, DOI: [10.1080/07370652.2011.569831](https://doi.org/10.1080/07370652.2011.569831)

To link to this article: <https://doi.org/10.1080/07370652.2011.569831>



Published online: 28 Mar 2012.



Submit your article to this journal [↗](#)



Article views: 1067



View related articles [↗](#)



Citing articles: 8 View citing articles [↗](#)

Time–Temperature Analysis, Tension and Compression in PBXs

DARLA GRAFF THOMPSON,
RACCI DELUCA, and GEOFF W. BROWN

High Explosives Science and Technology, Los Alamos
National Laboratory, Los Alamos, New Mexico

Time–temperature principles are applied to a wide range of mechanical properties data for two highly-filled polymer composites. Compressive and tensile properties were measured for plastic-bonded explosives (PBX) 9501 and 9502, spanning a wide range of strain rates and temperatures. Stress–strain parameters were determined and carried forward in a time–temperature analysis. For PBX 9501, best-fit constants were -8.0 and -10.0 K per decade of strain rate, for compression and tension, respectively. For PBX 9502, the best-fit constant for all data was -8.0 K/rate decade. Dynamic mechanical analysis (DMA) data for both formulations adhere to the same or similar time–temperature shift factors.

Keywords: DMA, mechanical properties, PBX 9501, PBX 9502, SHPB

Address correspondence to Darla Graff Thompson, WX-7, MS C920, High Explosives Science and Technology, P.O. Box 1663, Los Alamos National Laboratory, Los Alamos, NM 87545. E-mail: dkgraff@lanl.gov

Introduction

The mechanical response of a plastic-bonded explosive (PBX) composite is complex and dependent on a large number of parameters. Some parameters are easily controlled experimentally, such as temperature and strain rate [1–8], whereas others have been measured and correlated, such as specimen density [9]. Still other parameters such as storage humidity and pressing protocols have been purposefully varied and the effects evaluated [10,11]. Attempts to model and predict the mechanical response of PBX composites require an ability to mathematically describe the relevant experimentally observed dependencies. Any opportunity to demonstrate parameter interrelationships in determining the mechanical response of a PBX is potentially very useful in the development of a model with predictive capabilities.

The principles of time–temperature analysis were developed to demonstrate inherent correlations that exist in the mechanical response of polymers [12,13]. Specifically, if a polymer possesses a range of fast to slow relaxation times, and because the thermal environment may increase or decrease the rate of a response, it is reasonable to believe that decreased temperatures at slow displacement rates may involve the same relaxation processes as increased temperatures at fast rates. In recent years, a simplified time–temperature relationship was applied successfully to compressive strength data from two polymers, polycarbonate and polyvinylidene difluoride [1]. Subsequently, this simplified analysis was applied to the compression data of a PBX-type explosive, EDC 37 [2]. In addition to simply demonstrating that the mechanical response at high strain rates (and warm temperatures) versus the response at reduced temperatures (and low strain rates) is the same, this article demonstrates that one may predict, with some confidence, the mechanical response of PBX in particular regions of the compressive parameter space that are difficult to measure experimentally.

Microstructural effects associated with processing and damage of PBX 9501 and PBX 9502 have been studied and reported

by others [14–17]. Though the two composites each have 95 wt% loading of solid explosives, the similarity ends there. Particle size distribution and crystal morphology are entirely different for HMX and TATB, and the binder systems are vastly different as well (a heavily plasticized elastomer vs. a Teflon-like polymer). In mechanical model development and data analysis, juxtaposition of the two PBX compositions and responses is useful in highlighting similarities and differences between the two.

We present here the results of two extensive studies of both tensile and compressive properties for PBX 9501 and PBX 9502. Employing the simplified time-temperature relationship proposed by Siviour et al. [1], where changes in temperature are directly related to decade shifts in strain rate, we show that for each formulation the same (or similar) shift factors can be applied to both tension and compression, along with dynamic Split-Hopkinson pressure bar (SHPB) data, dynamic mechanical analysis (DMA) data, and tensile creep data. Our quasistatic analysis is conducted not only using strength data, as is typical in the time-temperature literature, but also includes modulus and failure strain values. In specific regions where mechanical parameters lack correlation, results are discussed in terms of specimen size effects and thermal transitions in the PBX binders. When time-temperature analysis results are corroborated over a large parameter space and using such a variety of different mechanical testing techniques, the principles may be applied with some confidence to aid in the development of predictive models.

Experimental

PBX 9501 and PBX 9502 Compositions and Processing

The PBX 9501 formulation contains approximately 95 wt% HMX (octahydro-1,3,5,7-tetranitro-1,3,5,7-tetrazocine) crystals in a bimodal size distribution. The nearly 5 wt% binder is a 1:1 mixture of Estane 5703 (Estane), a polyester polyurethane block copolymer, and a eutectic mix of

bis(2,2-dinitropropyl)acetal-formal (BDNPA-F, also called *nitroplasticizer*, NP). Approximately 0.1 wt% stabilizer is included.

PBX 9502 is comprised of 95 wt% triaminotrinitrobenzene (TATB) crystals and 5 wt% of Kel-F binder (a 3:1 chlorotrifluoroethylene : vinylidene fluoride copolymer). PBX 9502 is introduced in two forms, virgin and recycled. Only virgin PBX 9502 results are reported in this article, for which all of the ingredients were new and pristine; therefore, in this article, PBX 9502 refers only to the virgin material.

In typical PBX formulations, solid explosive particles are dispersed in water and slurry-mixed into the solvated polymeric binder. As the solvent is removed with heat, the polymer coats the particles and agglomerates to form molding powder or prills. After drying, the prills are compacted at high pressure to form high- and uniform-density charges that are precision machined into components or test specimens.

Quasistatic Uniaxial Specimens

Compression specimens were right circular cylinders. PBX 9501 cylinders were 0.953 cm in diameter and 2.858 cm long, and PBX 9502 cylinders were 0.953 cm in diameter and 1.905 cm long.

For both PBX 9501 and PBX 9502, tension specimens were cylindrical “dogbones” with $\sim 15^\circ$ tapered ends that seat, upon loading, into similarly tapered grip fixtures mounted in the load train. Due to the energetic nature of the material, no direct clamping or pinching of the specimens is allowed. Dogbones were 7.62 cm long, with a diameter of 2.154 cm at both ends. The gauge section of the dogbone was 3.81 cm long and 1.27 cm in diameter.

Quasistatic tension and compression specimens were machined from isostatically pressed hemispherical charges. PBX 9501 specimens, lot BAE 05G145-011, had an average pressed density of $1.8388 \pm 0.0013 \text{ g/cm}^3$ ($\sim 99\%$ of theoretical maximum density, TMD). PBX 9502 specimens, virgin lot HOL844890-022, had an average pressed density of $1.888 \pm 0.0020 \text{ g/cm}^3$ ($\sim 97\%$ TMD). In the machining process,

all specimens were cored such that their long axis was close to the inner surface and approximately tangential to it.

Quasistatic Uniaxial Tension and Compression Testing

Quasistatic uniaxial tension and compression tests were conducted remotely on an Instron 5567 (Instron Corp., Norwood, MA). Strain was measured using two oppositely mounted knife-edge extensometers. Tests were conducted in strain rate control using the averaged strain output of the two extensometers. Specimen temperature was controlled using a Bemco environmental chamber (Bemco, Inc., Simi Valley, CA, USA), with thermal ramps of ~ 1 K/min and ~ 20 min of soak time once the target temperature was achieved. Under these conditions, thermal gradients at the interior of mock PBX specimens were measured as less than 1 K. Test temperatures were 258, 273, 296, 313, and 323 K for PBX 9501 and 221, 253, 293, 323, and 347 K for PBX 9502. In compression, the physical manifestation of specimen failure is relatively subtle until strain values well past the stress maximum are achieved. For PBX 9501 ($l/d=3$), compression specimens were observed to barrel near the center of the cylinder, but fracture did not occur. For PBX 9502 ($l/d=2$), at high strains, compression specimens almost always broke into two pieces with a $\sim 60^\circ$ fracture line. All tensile dogbones failed abruptly, even at warm temperatures, with a fracture surface perpendicular to the long axis. No tensile necking was observed.

For the analysis, stress-strain curves were characterized using three parameters [18]. Ultimate stress, σ_m , is the highest stress value achieved before failure of the material; ultimate strain, ε_m , is the strain value corresponding to σ_m ; the modulus, E_{25} , is the slope of the linear elastic portion of the stress-strain curve at 25% of σ_m . Due to the seating of the specimen under load, the low-strain region of the stress-strain curve can sometimes be affected by an experimental “toe” artifact. A toe-corrected ε_m value was determined by calculating the strain intercept value at zero stress for the E_{25} tangent line and then

shifting the strain data by this amount so that the tangent line passed through the origin.

The number of quasistatic specimens tested at each experimental condition ranged from three to five. The average value of each stress-strain parameter was calculated and used in the plots and analyses presented in this article. The standard deviation is shown in error bars (Figures 1, 3, 7 and 8). Of all of the parameters, ultimate strain showed the largest statistical variation. In most cases, for the other parameters, error bars are masked by the physical size of the plot symbol used.

Split-Hopkinson Pressure Bar Measurements

Dynamic SHPB tests were conducted using a gas gun assembly with either Ti-6Al-4V or magnesium AZ31B-alloy bars, appropriate for the evaluation of low-strength materials [5–8]. Specimen dimensions were 6.35 mm diameter by 6.35 or 3.15 mm long. Specimen temperature was controlled using a gas manifold system and was varied between 218 and 328 K. The SHPB PBX 9501 data used in this analysis are summarized in Fig. 2 of Williamson et al. [2]. Virgin PBX 9502 specimens were machined from hydrostatically pressed billets with densities just slightly lower than nominal [8, 19], approximately 1.885 g/cm³.

Dynamic Mechanical Analysis

DMA measurements were conducted using a Model 2980 dynamic mechanical analyzer (TA Instruments, New Castle, DE, USA) with a single cantilever clamp [20,21]. At each step in temperature between –173 and 373 K, the method applies a 20- μ m amplitude displacement at five frequencies. Rectangular specimens, 10 $\times$ 24 $\times$ 2 mm, were machined from hydrostatically pressed billets.

Uniaxial Tensile Creep

Short-term (<3 days) tensile creep tests were conducted on PBX 9501 using the Instron 5567 workstation and the dogbone

specimen geometry described above. Specimens were loaded in tension using a constant crosshead speed of 0.254 mm/min and held at load until failure occurred, with oppositely mounted knife-edge extensometers recording strain as a function of time. The Bemco chamber, described above, was used to hold test temperatures at 273, 294, 299, and 305 K during the duration of the test. Several time-to-failure tests were performed in the absence of extensometers, confirming that the presence of extensometers did not always lead to shorter test durations. For these tests, the stress maximum was taken to be the constant load per specimen area, and the strain rate was obtained using a linear fit to the secondary (linear) part of the creep curve, after the initial loading in the elastic range of the material and before any indication of specimen failure. Specimens were obtained from nine hemispherical PBX 9501 charges representing different lots.

Time-Temperature Superposition Analysis

In our analysis of PBX 9501 and PBX 9502 data, we variously considered three time-temperature relationships from the literature. Each one attempted to relate the logarithmic dependence of a time-dependent parameter to test temperature using one or two empirical constants. This relation is expressed as a shift function $\log(a_T)$, which is a function of temperature. In Eq. (1), from Glasstone et al. [12], the change in frequency ν , and test temperature are related by a constant, E_a , interpreted as an activation energy:

$$\log a_T = \log\left(\frac{\nu_1}{\nu_2}\right) = \frac{-E_a}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right) \quad (1)$$

Similarly, the well-known Williams-Landel-Ferry (WLF) equation relates the change in η , material viscosity, to the test temperature through two constants [13]:

$$\log a_T = \log\left(\frac{\eta}{\eta_{T_s}}\right) = \frac{-C_1(T - T_s)}{C_2 + (T - T_s)} \quad (2)$$

Though the functional form and theoretical basis for these two equations are different, they both employ concepts of time-based relaxation processes in the material. Finally, in the simplified relationship applied to polymers by Siviour et al. [1], the change in applied strain rate, $\dot{\epsilon}$, is related to differences in test temperature using a single constant, A :

$$\log a_T = \log \left(\frac{\dot{\epsilon}_{\text{ref}}}{\dot{\epsilon}} \right) = \frac{1}{A} (T_{\text{ref}} - T) \quad (3)$$

In an application of Eq. (3) to the quasistatic and dynamic compression results of a PBX-type material, EDC 37, it was determined that $A = -13$ K per decade of strain rate [2].

In our present analysis of existing PBX 9501 and PBX 9502 mechanical properties data, we report only the results obtained using the simplified relationship, Eq. (3). We applied Eqs. (1) and (2) to our data for comparison, usually seeking to better capture subtleties in the material behavior. These results will be mentioned in passing, but they will not be formally presented.

Results and Discussion

PBX 9501: Compression, Tension, SHPB, Tensile Creep

In Fig. 1a all of the values of each of the three quasistatic compression parameters for PBX 9501 versus strain rate are plotted, with the temperature shown in the legend. These data demonstrate the expected trend that stress and modulus values increase with increasing strain rate and decreasing temperature. It has previously been stated [2,18] that PBX 9501 compressive failure occurs with relatively consistent ultimate strain values, and we point out in Fig. 1a (center) that for PBX 9501 the strain values were observed to vary in a statistically significant way as a function of strain rate and temperature.

In Fig. 1b the same three compression parameter values as in Fig. 1a are shown but now shifted on the strain-rate axis using Eq. (3), with $A = -8$ K per decade of strain rate, to an

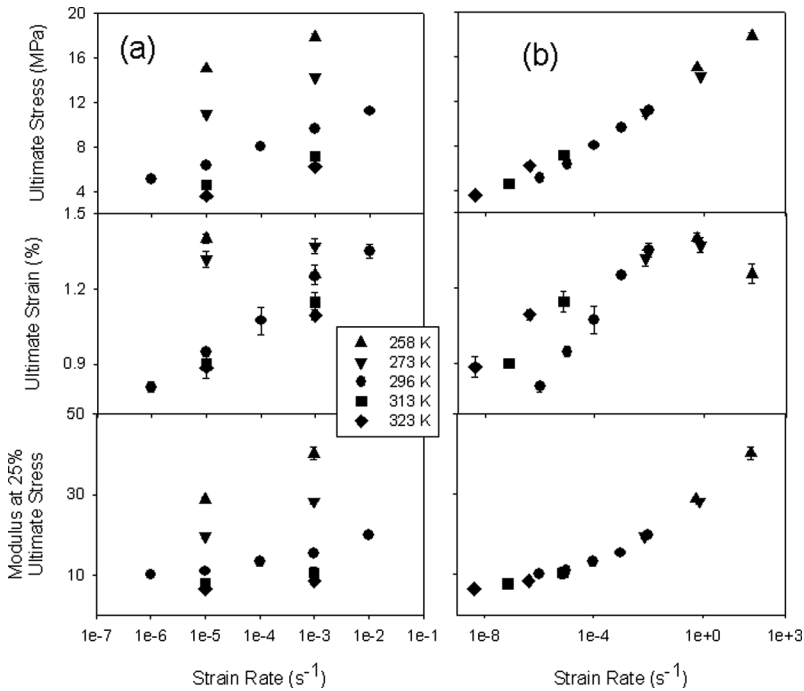


Figure 1. PBX 9501 (a) as-measured and (b) time–temperature-shifted values of three quasistatic compressive parameters are plotted versus strain rate. Ultimate stress (top), ultimate strain (middle), and modulus (bottom) parameters are taken from stress–strain curves as defined in the text. Test temperatures are defined in the legend.

equivalent test temperature of 23°C (296 K). The shifted stress and modulus values (Fig. 1b, top and bottom) align to form continuous and consistent master curves. In addition, the shifted ultimate strain values (1b, center) form a consistent master curve below room temperature but not above. It is possible that this change is due to a difference in failure mechanism or more likely is related to phase mixing in the binder as the temperature exceeds ambient.

The Estane binder used in PBX 9501 is a polyester–urethane elastomer. The polymer exists in a state of hydrogen-bond

equilibrium between a hard domain (hydrogen-bond donor and acceptors from urethane groups) and a soft domain (ester groups, hydrogen-bond donors only). Hard domains act like physical cross-links, providing strength to the material. As the temperature is increased above ambient, the number of Estane hard domains is known to be affected as the two domains begin to phase-mix [22]. Though no specific mechanism is proposed here, phase mixing in polyurethane–polyester has been shown to degrade the mechanical integrity of the polymer. The loss of a consistent master curve for the ultimate strain values measured above ambient (~ 298 K) may well be the result of Estane phase mixing; in addition, the melt temperature of NP is about 288 K [23,24], which could play a role. It is not clear why stress and modulus values are not similarly affected.

In Fig. 2 the shifted ultimate stress values are replotted from Fig 1b (top) along with the shifted (296 K equivalence) SHPB

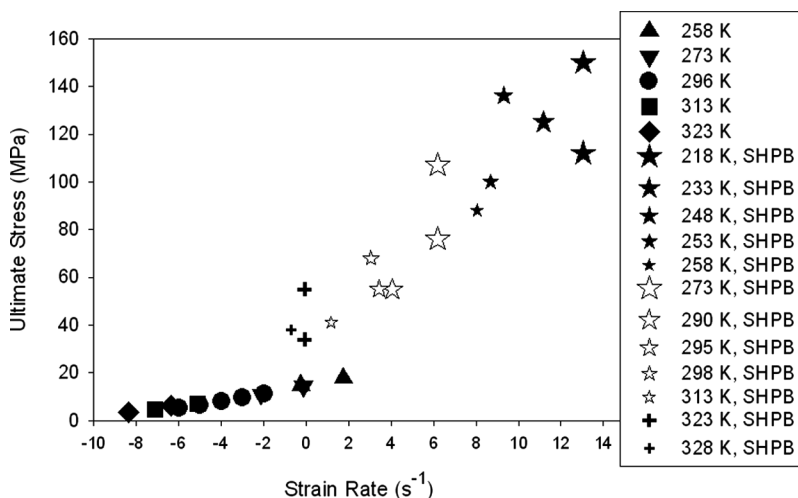


Figure 2. PBX 9501 ultimate stress values from Fig. 1b (top) are plotted with time–temperature-shifted SHPB stress values (see text for SHPB references). Legend denotes test type and temperature.

stress values (for unshifted values, plotted vs. temperature, see Fig. 2 in Williamson et al. [2]). Though a continuous master curve is suggested, a clear vertical discontinuity is observed between the quasistatic and SHPB results, with possibly a slope change as well. At high strain rates for EDC37, Williamson et al. [2] observed a sharp increase in the slope of stress versus strain rate; however, they did not observe the vertical shift or discontinuity between quasistatic and dynamic data. We initially believed that the primary cause of the vertical discontinuity could be the result of the small aspect ratio for the SHPB specimens (higher confinement of ends). However, quasistatic tests on SHPB specimens have accounted for only $\sim 15\%$ of the vertical shift. It has been proposed (Alex Zubelewicz, LANL, personal communication) that the fracture propagation speed of PBX 9501 may be slower than the SHPB impact speed, so that the apparent material strength at high-rate impact is significantly higher.

Figure 3a contains plots of the three PBX 9501 tensile parameters obtained from a large series of quasistatic tests (filled symbols). The general dependence on temperature and strain rate is nearly identical to that observed in compression (Fig. 1a). Note that for this material, at the same temperature and quasistatic strain rate, the tensile modulus is a factor of ~ 2 higher than the compressive. Additionally, the ultimate tensile stress and strain are approximately 25 and 30%, respectively, of the compressive values. Figure 3a (top) also shows the ultimate stress values versus strain rate obtained from a small series of tensile creep tests. These data are designated with open symbols.

In Fig. 3b, the time-temperature-shifted tensile parameter values from Fig. 3a are all plotted using Eq. (3) and $A = -10$ K per decade of strain rate, shifted to an equivalent test temperature of 298 K. The aligned master curve for ultimate stress is excellent, including both quasistatic and creep data. The modulus master curve shows a small discontinuity between ambient and low temperatures. The shifted ultimate strain data show trends relating strain rate and temperature but do not overlay to form a master curve.

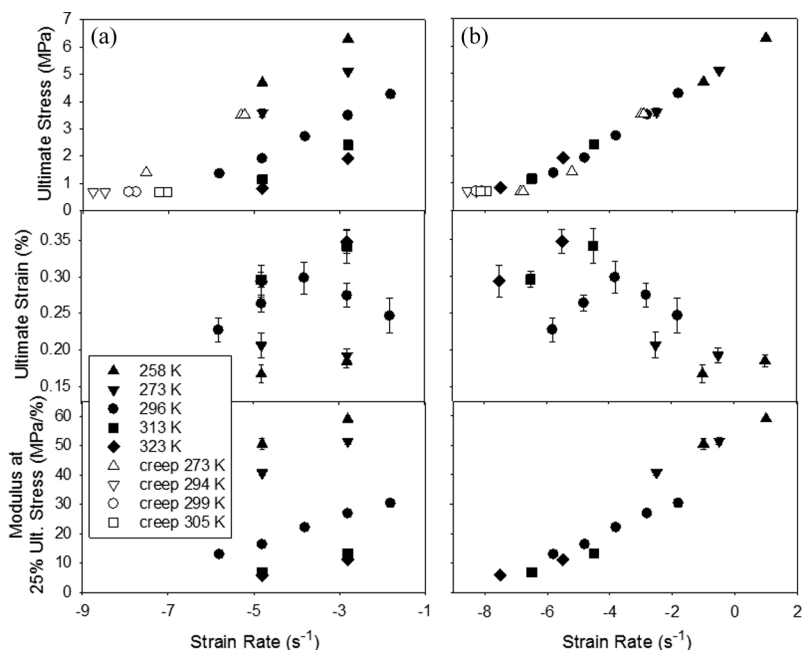


Figure 3. PBX 9501 (a) as-measured and (b) time-temperature-shifted values of three quasistatic tensile parameters are plotted versus strain rate. Also included with the quasistatic ultimate stress values (top) are ultimate stress values from tensile creep tests (open symbols; see text). Test types and temperatures are defined in the legend.

PBX 9501: DMA

In the simplest approximation, the single cantilever beam DMA test configuration provides the in-phase (storage) and out-of-phase (loss) elastic (or Young's) modulus as a function of the displacement frequency and temperature. A small amplitude displacement is used throughout the test, so that the elastic limit of the material is not exceeded. It is fully expected that DMA data for PBX 9501 should follow similar time-temperature superposition relationships as observed above for tension and compression, and previous work has been published [24,25] on torsional DMA results for PBX 9501 and its binder.

Though DMA mechanical response is often plotted on a temperature axis, in Fig. 4a we plotted the storage modulus and the tan delta (loss/storage) response at different temperatures (see legend) as a function of frequency. In Fig. 4b, the same storage modulus and tan delta values are plotted on a frequency-shifted axis for a temperature equivalence of 296 K. The shifts were performed using Eq. (3) with $A = -8$ K per decade of frequency, the same used above for the compression data, and resulted in satisfactory master curves. Alignment in the master curves of Fig. 4b can be slightly improved using Glasstone et al.'s [12] relation (Eq. (1)), with $E_a = 85$ kJ/mol (data not shown). Glasstone et al.'s relation allows some curvature in the relationship between the shift factor and the temperature change, as seen in Fig. 5 (slightly larger $\log(a_T)$ values at temperature extremes), and this helps the alignment at high and low temperatures (shifted data for Glasstone et al.'s vs. Siviour

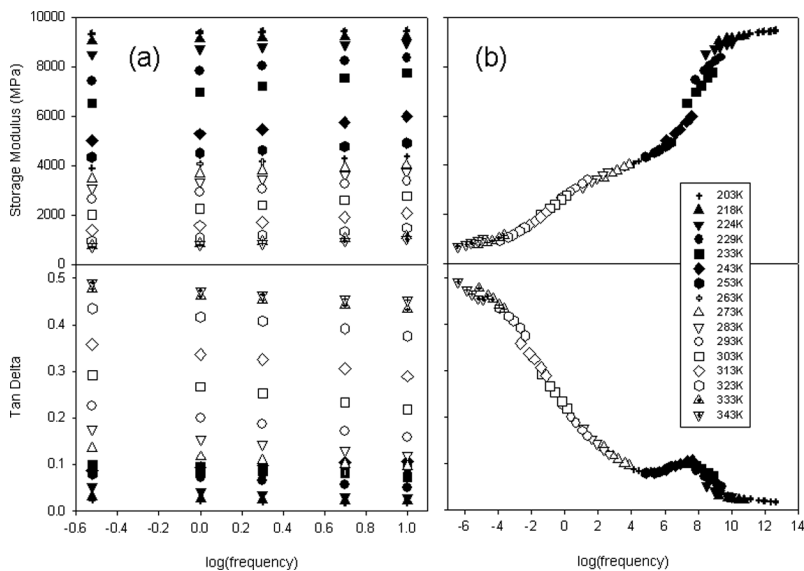


Figure 4. PBX 9501 DMA results: (a) as-measured and (b) time-temperature-shifted values of storage modulus (top) and tan delta (bottom). Temperatures are listed in the legend.

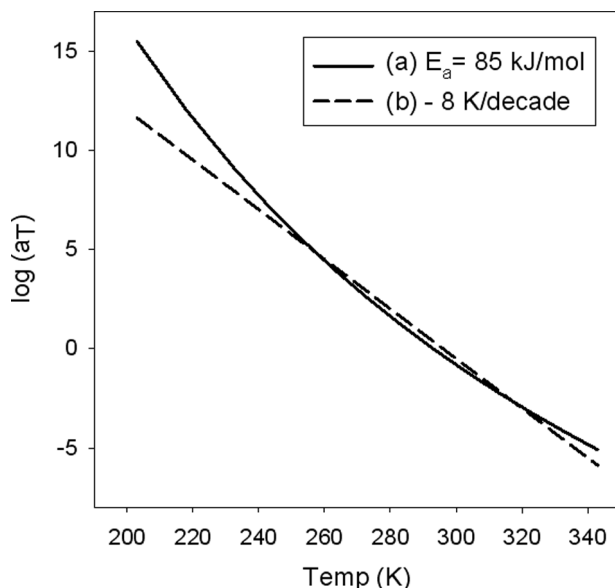


Figure 5. Plot comparing best-fit $\log a_T$ versus temperature ($T_{\text{ref}}=296\text{ K}$) for Glasstone et al.'s [12] equation (Eq. (1)) and Siviour et al.'s [1] relation (Eq. (3)). Only results from Siviour et al.'s relation are used in the analysis (see Fig. 4b).

et al.'s relations are included as supplemental information in Fig. A1).

In addition to the creation and evaluation of master curves using shift factors, another useful means of determining the adherence of DMA data to the principles of time-temperature superposition is to make a Wicket or Cole-Cole plot, as shown in Fig. 6 [26]. Here, the measured $\tan \delta$ values are plotted versus the storage modulus (i.e., no shifting is involved). If the plotted data form a curve in the shape of an arch (or a croquette wicket!) then the assumption of one relaxation time through the data set is validated. In Fig. 6 a wicket shape is observed for the data at or below approximately 263 K. The glass transition, T_g , of the Estane/NP binder as measured by modulated differential scanning calorimetry (MDSC) was 225 K, consistent with the idea of a single dominant relaxation

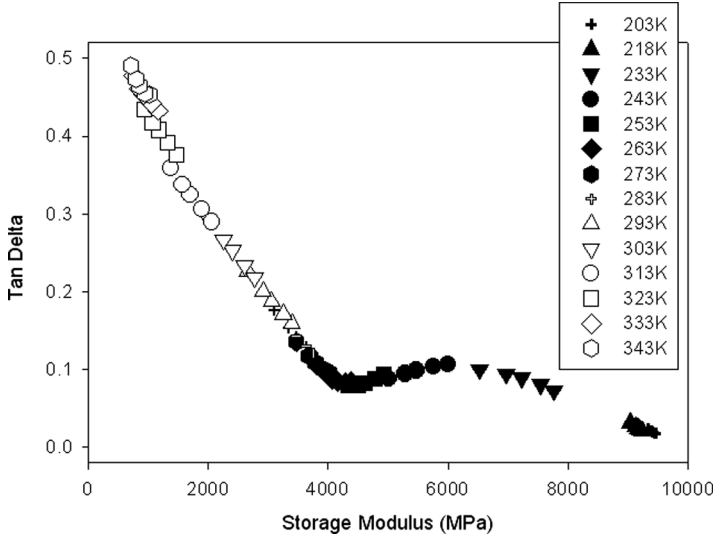


Figure 6. The Cole-Cole or Wicket plot for PBX 9501 DMA data.

time for low temperatures [24]. As mentioned, NP melts around 288 K and the H-bonded cross-links of Estane hard domains are known to be disrupted significantly as the temperature exceeds ambient. These factors, along with the cantilever configuration used in our measurements, may be the cause of the limited validity range we observed in our Wicket plot analysis. Earlier dynamic torsional (shear) characterization of the Estane/NP binder showed time–temperature validity from about 223 to 313 K [25].

Virgin PBX 9502: Compression, Tension, SHPB

The quasistatic and dynamic (SHPB) compression parameters for virgin PBX 9502 are plotted in Fig. 7a. For these data, three specimens were tested at each condition, but every combination of temperature and strain rate was tested. Using Eq. (3) and $A = -8$ K per decade of strain rate, the data in Fig. 7a were shifted to an equivalent testing temperature of 293 K, plotted

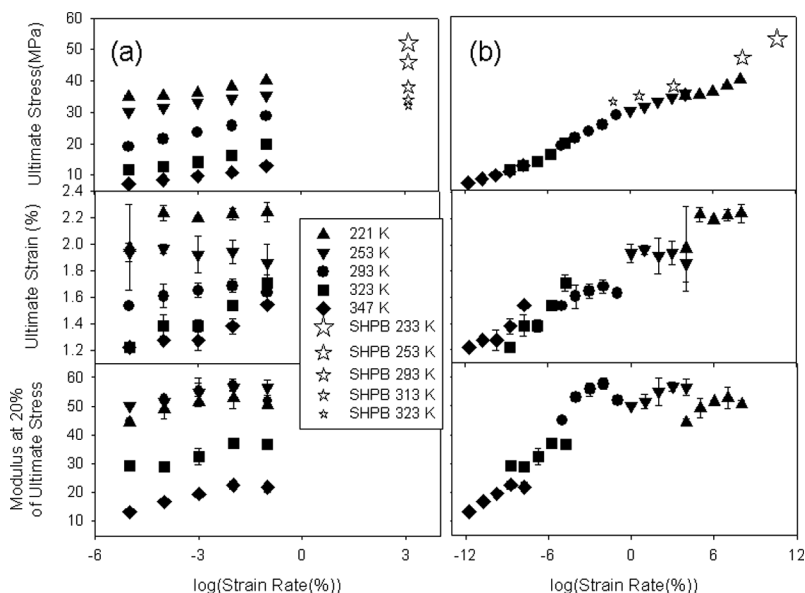


Figure 7. PBX 9502 (a) as-measured and (b) time-temperature-shifted values of three quasistatic compressive parameters are plotted versus strain rate. SHPB ultimate stress values are included (open stars, top). Compressive parameters are taken from stress-strain curves as defined in the text. Test types and temperatures are defined in the legend.

in Fig. 7b. Alignment of the ultimate stress (top) master curve is excellent, and the ultimate strain and modulus master curves (middle and bottom) are well-aligned and contiguous as well, showing better consistency with time-temperature principles than observed for PBX 9501. Note that for two very different PBX materials, the best-fit shift factor was the same.

The PBX 9502 ultimate stress master curve shows very good agreement of the SHPB and quasistatic values, as opposed to the vertical offset observed in Fig. 2 for PBX 9501. We explored the possible role of specimen density on the measured ultimate stress values; previous work indicated that this could be a factor [9,11], but the data showed no density correlation. It may be that the fracture propagation speed is faster in PBX 9502 than

in PBX 9501, providing a smoother and more consistent strength response as the strain rate is increased to SHPB levels.

The three quasistatic tension parameters are plotted versus strain rate in Fig. 8a. The master curves at an equivalent test temperature of 293 K are shown in Fig. 8b, using $A = -8$ K per decade strain rate in Eq. (3). As observed for the PBX 9502 compression, the master curve alignment for all three parameters is quite good. The ultimate strain value does not change appreciably over the parameter space and does not show the inconsistencies that were seen in the shifted PBX 9501 tensile strain values.

Virgin PBX 9502: DMA

Figure 9a shows the as-measured DMA response for the PBX 9502 storage modulus (top) and tan delta (bottom). Figure 9b

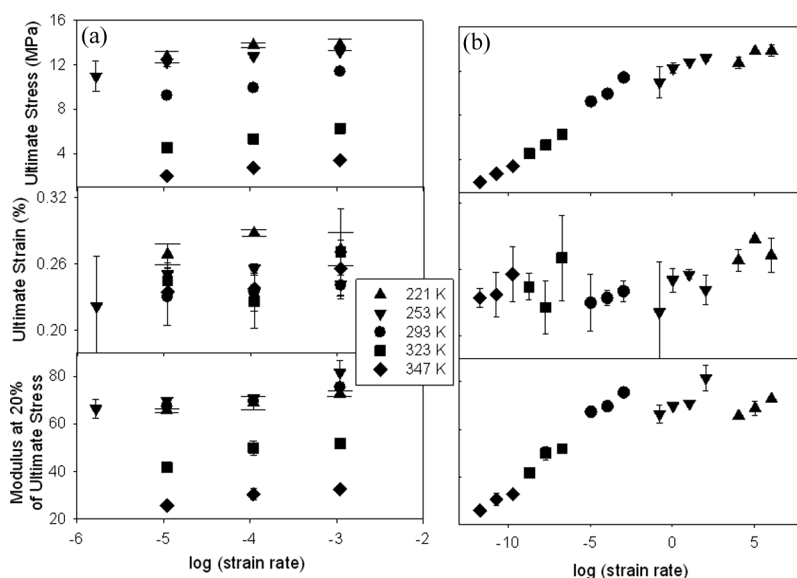


Figure 8. PBX 9502 (a) as-measured and (b) time–temperature-shifted values of three quasistatic tensile parameters are plotted versus strain rate. Temperatures are given in the legend.

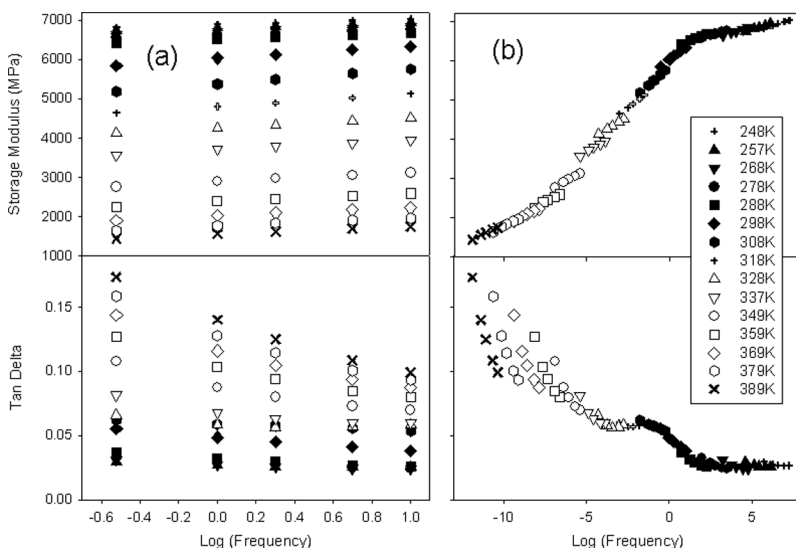


Figure 9. PBX 9502 DMA results: (a) as-measured and (b) time-temperature-shifted values of storage modulus (top) and tan delta (bottom). Temperatures are listed in the legend.

shows the same values shifted to an equivalent temperature of 296 K using Eq. (3) and -8 K/decade of strain rate, the same as used in the shift of compression and tension parameters for this material. The master curve alignment is good for the storage modulus. However, similar to what was observed for the PBX 9501 DMA, the tan delta master curve alignment shows deviations at warm temperatures. For PBX 9502, master curve alignment is lost above about 328 K where crystalline Kel-F is known to undergo melting [27,28].

A comparison of Wicket plots for PBX 9501 and PBX 9502 (Figs. 6 and 10, respectively) shows an interesting similarity. Both plots exhibit the wicket-like appearance in the low-temperature region, with deviation occurring at approximately $T_g + 30$ K (approximately 328 K for PBX 9502 and 263 K for PBX 9501). The two different polymers are known to undergo unique physical changes at these temperatures, and the observations are consistent with an upper temperature limit for

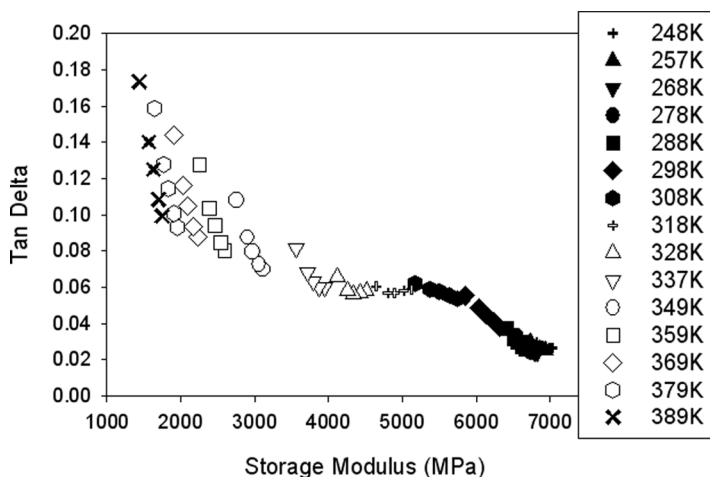


Figure 10. Cole-Cole or Wicket plot, PBX 9502 DMA data.

the single relaxation time that dominates the region of parameter space being probed in the DMA measurement. As demonstrated by Hoffman [28], Kel-F crystallinity can be removed by preheating to 393 K. Specimens treated in this way do not show the Kel-F melt transition and their DMA time-temperature agreement extends to temperatures as high as 353 K.

Conclusions

A range of techniques has been used to measure the mechanical response of two very different PBX compositions. Using principles of time-temperature superposition, the interrelationships of these different data sets were demonstrated and quantified. To our knowledge, this constitutes the first time-temperature analysis of PBX composites that has included not only ultimate stress values from quasistatic data but also ultimate strain and modulus values. In addition, it is the first comprehensive comparison of quasistatic tensile and compressive response. We have demonstrated not only good agreement in the measurements

from different experimental techniques, but also consistent shift factors from tension to compression for a given material.

Though shift factors were observed to be (coincidentally) very similar for PBX 9501 and PBX 9502 data sets, the trends observed were entirely consistent with the markedly different T_g values of the two binder systems. Specifically, for the two materials we showed that for temperatures lower than approximately $T_g + 30$ K, the principles of time-temperature superposition are valid for the composites at DMA frequencies; above these respective temperatures, the two polymeric binders undergo transitions unique to each that relate to crystallinity and phase segregation and no longer adhere to time-temperature principles. For quasistatic tension and compression, the time-temperature-shifted PBX 9501 ultimate strain values over the temperature range do not align well to form a master curve. The shifted values in compression (Fig. 1b, middle), but particularly in tension (Fig. 3b, middle), show that at high strain rates, a reduction in rate results in a higher ultimate strain; at low rates, a further reduction in rate results in a lower ultimate strain. We believe that this difference indicates a change in the PBX 9501 failure mechanism as one traverses the temperature/rate range. It may also derive from the hard-soft domain mixing that occurs in the Estane binder of PBX 9501 as the temperature is increased above ambient. For PBX 9502, the shifted stress, strain, and modulus values all align well to form consistent master curves for each.

Our PBX 9501 data show that high-rate tests give much higher ultimate stresses than what would be predicted by equivalent quasistatic tests using time-temperature shifts. Additional quasistatic testing on PBX 9501 showed that the small specimen size and low aspect ratio of high-rate specimens can account for only $\sim 15\%$ of the stress increase observed in the dynamic tests. For EDC 37 [2] and PBX 9502, the equivalent quasistatic and dynamic ultimate stress values agree very well and form a contiguous master curve. We hypothesize that the rate of damage accumulation in PBX 9501 is slower than the rate of the dynamic SHPB tests, giving rise to a high and

transient stress state. Detailed exploration of the intermediate strain rate region of PBX 9501 compression would aid greatly in understanding this unique strain-rate dependence of the ultimate stress values.

Our primary interest in the time–temperature analysis presented here is to gain confidence in the interpolation and short-range extrapolation of material response through parameter space that is not directly measured. In addition, it is always desirable to show corroborative information collected from various experimental sources and to see consistency in tension versus compression tests. Model development efforts are ongoing [29–35] to describe the mechanical behavior of PBX over a wide range of temperatures and strain rates toward the exercise and application of predictive capabilities. To the extent that time–temperature principles are observed in material response, significant aid is generated for the mathematical modeling effort.

Acknowledgments

Funding for this work was provided by LANL Campaign 8 (Enhanced Surveillance) under direction of Sheldon Larson and Thomas Zocco. DMA data were obtained by Mary S. Campbell. LANL is operated by LANS, LLC, under DOE/NNSA contract DE-AC52-06NA25396.

References

- [1] Siviour, C. R., S. M. Walley, W. G. Proud, and J. E. Field. 1984. The high strain rate compressive behaviour of polycarbonate and polyvinylidene difluoride. *Polymer*, 46: 12546–12555.
- [2] Williamson, D. M., C. R. Siviour, W. G. Proud, S. J.P. Palmer, R. Govier, K. Ellis, P. Blackwell, and C. Leppard. 2008. Temperature–time response of a polymer bonded explosive in compression (EDC 37). *Journal of Physics D: Applied Physics*, 41: 85404–85414.
- [3] Funk, D. J., G. W. Laabs, P. D. Peterson, and B. W. Asay. 1995. Measurement of the stress/strain response of energetic materials as a function of strain rate and temperature: PBX 9501 and

- Mock 9501. *Proceedings of APS Shock Compression of Condensed Matter*, August 13–18, Seattle, WA.
- [4] Idar, D. J., P. D. Peterson, P. D. Scott, and D. J. Funk. 1997. Low strain rate compression measurements of PBXN-9, PBX 9501 and Mock 9501. *Proceedings of APS Shock Compression of Condensed Matter*, July 27–August 1, Amherst, MA.
- [5] Gray, G. T., III, W. R. Blumenthal, D. J. Idar, and C. M. Cady. 1997. Influence of temperature on the high-strain rate mechanical behavior of PBX 9501. *Proceedings of APS Shock Compression of Condensed Matter*, July 27–August 1, Amherst, MA.
- [6] Gray, G. T., III, D. J. Idar, W. R. Blumenthal, C. M. Cady, and P. D. Peterson. 2000. High- and low-strain rate compression properties of several energetic material composites as a function of strain rate and temperature. *Proceedings of the 11th International Detonation Symposium*, 30 August–September 4, Snowmass Village, CO.
- [7] Idar, D. J., D. G. Thompson, G. T. Gray III, W. R. Blumenthal, C. M. Cady, P. D. Peterson, E. L. Roemer, W. J. Wright, and B. L. Jacquez. 2001. Influence of polymer molecular weight, temperature and strain rate on the mechanical properties of PBX 9501. *Proceedings of APS Shock Compression of Condensed Matter*, June 24–29, Atlanta, GA.
- [8] Blumenthal, W. R., G. T. Gray III, D. J. Idar, M. D. Holmes, P. D. Scott, C. M. Cady, and D. D. Cannon. 1999. Influence of temperature and strain rate on the mechanical behavior of PBX 9502 and Kel-F 800. *Proceedings of APS Shock Compression of Condensed Matter*, June 27–July 2, Snowbird, UT.
- [9] Meyer, T., R. V. Browning, and R. Spence. 2000. *Compression properties of PBX 9501 at densities from 1.60 to 1.83 g/cc*. Amarillo, TX: Mason & Hanger Pantex Plant. Report DOE/AL 65030-00-04.
- [10] Browning, R. V. and T. Meyer. 2000. Effects of humidity and other variables on the strength of PBX 9501. *Proceedings of the 23rd Aging, Compatibility and Stockpile Stewardship Conference*, November 14–16, Livermore, CA.
- [11] Thompson, D. G., B. Olinger, and R. DeLuca. 2005. The effect of compaction parameters on the mechanical properties of plastic bonded explosives. *Propellants, Explosives, Pyrotechnics*, 30: 391–396.

- [12] Glasstone, S., K. J. Laidler, and H. Eyring. 1941. *The Theory of Rate Processes*. New York: McGraw-Hill.
- [13] Williams, M. L., R. F. Landel, and J. D. Ferry. 1955. The temperature dependence of relaxation mechanisms in amorphous polymers and other glass forming liquids. *Journal of the American Chemical Society*, 77: 3701–3707.
- [14] Skidmore, C. B., D. S. Phillips, S. F. Son, and B. W. Asay. 1997. *Characterization of HMX Particles in PBX 9501*. Los Alamos, NM: Los Alamos National Laboratory. LANL report LA-UR 97-2596.
- [15] Skidmore, C. B., D. S. Phillips, B. W. Asay, D. J. Idar, P. M. Howe, and D. S. Bolme. 1999. *Microstructural effects in PBX 9501 damaged by shear impact*. Los Alamos, NM: Los Alamos National Laboratory. LANL report LA-UR 99-3233.
- [16] Peterson, P. D. and D. J. Idar. 2005. Microstructural difference between virgin and recycled lots of PBX 9502. *Propellants, Explosives, Pyrotechnics*, 30: 88–94.
- [17] Rae, P. J., S. J.P. Palmer, H. T. Goldrein, J. E. Field, and A. L. Lewis. 2002. Quasi-static studies of the deformation and failure of PBX 9501. *Proceedings of the Royal Society of London A*, 458: 2227–2242.
- [18] Wiegand, D. A. 1997. Mechanical failure properties of composite plastic bonded explosives. *Proceedings of APS Shock Compression of Condensed Matter*, July 27–August 1, Amherst, MA.
- [19] Thompson, D. G., G. T. Gray III, W. R. Blumenthal, C. M. Cady, W. J. Wright, and B. Jacquez. 2002. Quasi-static and dynamic mechanical properties testing of PBX 9502: Strain rate, temperature, density and processing methods. *Proceedings of Life Cycles of Energetic Materials*, July 14–17, Westminster, CO.
- [20] Garcia, D. A. and M. S. Campbell. 2000. A comparison of the glass transition temperatures of PBX 9501 library lots with virtually-aged PBX 9501. *Proceedings of the 23rd Aging, Compatibility and Stockpile Stewardship Conference*, November 14–16, Livermore, CA.
- [21] Campbell, M. S. 2003. Dynamic mechanical analysis of plastic bonded explosives. *Proceedings of the 25th Aging, Compatibility and Stockpile Stewardship Conference*, November 18–20, Livermore CA.
- [22] Lee, H. S. and S. L. Hsu. 1989. An analysis of phase separation kinetics of model polyurethanes. *Macromolecules*, 22: 1100–1111.

- [23] Rivera, T. 1973. *Study to determine the phase diagram of the BDNPA/F system*. Los Alamos, NJ: Los Alamos National Laboratory. LASL Report LA-5313.
- [24] Hoffman, D. M. 2003. Dynamic mechanical signatures of a polyester-urethane and plastic bonded explosives based on this polymer. *Journal of Applied Polymer Science*, 83: 1009–1024.
- [25] Flowers, G. L. 1985. *Characterization of binders used in high explosives and the effect of binder strength on impact sensitivity of explosives*. Amarillo, TX: Mason & Hanger Pantex Plant. Report MHSMP-85-42.
- [26] Menard, K. P. 2008. *Dynamic Mechanical Analysis, A Practical Introduction*, 2nd ed. Boca Raton, FL: CRC Press.
- [27] Hoffman, D. M., F. M. Matthews, and C. O. Pruneda. 1989. Dynamic mechanical and thermal analysis of crystallinity development in Kelf-F 800 and TATB/Kel-F 800 plastic bonded explosives. Part I. Kel-F 800. *Thermochimica Acta*, 156: 365–372.
- [28] Hoffman, D. M. 2001. Dynamic mechanical signatures of aged LX-17-1 plastic bonded explosive. *Journal of Energetic Materials*, 19: 163–193.
- [29] Bennett, J. G., K. S. Haberman, J. N. Johnson, B. W. Asay, and B. F. Henson. 1998. A constitutive model for the non-shock ignition and mechanical response of high explosives. *Journal of the Mechanics and Physics of Solids*, 46: 2303–2322.
- [30] Hackett, R. M. and J. G. Bennett. 2000. An implicit finite element material model for energetic particulate composite materials. *International Journal for Numerical Methods in Engineering*, 49: 1191–1209.
- [31] Clements, B. F. and E. M. Mas. 2004. A theory for plastic bonded materials with a bimodal size distribution of filler particles. *Modelling and Simulation in Materials Science and Engineering*, 12: 407–421.
- [32] Rangaswamy, P., D. G. Thompson, C. Liu, and M. W. Lewis. 2010. Modeling the mechanical response of PBX 9501. *Proceedings of 14th International Detonation Symposium*, April 11–16, Coeur d’Alene, ID.
- [33] Tan, H., C. Liu, Y. Huang, and P. H. Geubelle. 2005. The cohesive law for the particle/matrix interfaces in high explosives. *Journal of the Mechanics and Physics of Solids*, 53: 1892–1917.
- [34] Tan, H., Y. Huang, and C. Liu. 2008. The viscoelastic composite with interface debonding. *Composites Science and Technology*, 68: 3145–3149.

- [35] Yan-Qing, W. and H. Feng-Lei. 2009. A micromechanical model for predicting combined damage of particles and interface debonding in PBX explosives. *Mechanics of Materials*, 41: 227–247.

Appendix

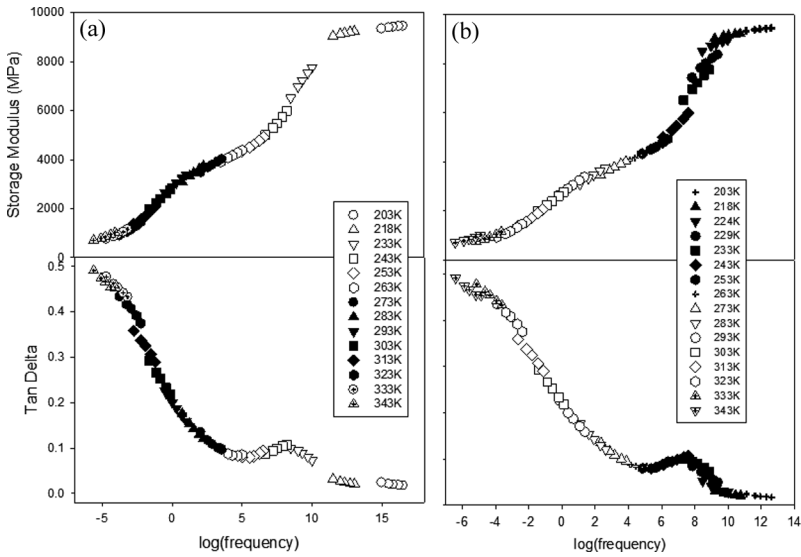


Figure A1. Shifted PBX 9501 DMA data using (left) Glasstone et al.'s [12] relation (Eq. (1)) and (right) Siviour et al.'s [1] relation (Eq. (2)) with parameters as shown.